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CST 395 Neural Network and Deep learning question paper

Neural Network and Deep learning (Dr. A.P.J. Abdul Kalam Technical University)

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree (Honours) Examination December 2023 (2021 Admission)

Course Code: CST 395

Course Name: NEURAL NETWORKS AND DEEP LEARNING

Max. Marks: 100 Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks) Marks

1 State the difference between overfitting and underfitting. (3)

2 Is this following task -a classification or prediction task? Justify (3)

"An emergency room in a hospital measures 10 variables like BP,sugar, age etc of newly admitted patients.A decision has to be taken whether to put the patient in ICU.Due to the high cost of ICU, those patients who may survive more than a month are given higher priority. The task is to predict high risk patients and discriminate them from low risk patients"

3 Explain any three activation functions used in neural networks. (3)

4 Explain a simple feed forward neural network with diagram (3)

5 Explain why Initializing the weights of a neural network with very small or large random numbers is not preferred. (3)

6 Discuss the importance of Drop out layer in neural networks. (3)

7 Differentiate between same convolution and valid convolution. (3)

8 Illustrate max pooling on the following input by a 2*2 kernel with stride 2 (3)

8	5	3	9	7	2
1	5	7	8	3	2
6	3	8	2	4	9
4	7	6	7	2	1
9	8	7	4	3	2
4	7	4	2	6	1

9 What are the main difficulties in training a RNN. (3)

10 Why a normal feed forward neural network cannot be used for text auto completion tasks. Which type of neural networks can be used for this type of tasks. (3)

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PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

11 a) What is a Confusion matrix. Define the terms Accuracy, Precision, Sensitivity and Specificity. (9)

b) Consider a 2 class classification problem of whether a person is diabetic or not. Let the data set consist of 10 records with the expected outcomes and the set of predicted outcomes by the classification algorithm. Draw the confusion matrix and compute the Accuracy, Precision and Sensitivity. (5)

No.	Actual	Predicted
1	Diabetic	Diabetic
2	Not Diabetic	Not Diabetic
3	Not Diabetic	Diabetic
4	Diabetic	Not Diabetic
5	Diabetic	Diabetic
6	Not Diabetic	Not Diabetic
7	Diabetic	Not Diabetic
8	Not Diabetic	Diabetic
9	Not Diabetic	Diabetic
10	Not Diabetic	Not Diabetic

12 a) Define machine learning. What is the significance of Loss functions in Machine Learning. List out any four applications of machine learning. (8)

b) Discuss different learning approaches used in machine learning. (6)

Module -2

13 a) Explain the problem of exploding and vanishing gradient problems while training a neural network. (8)

b) Assume you have a 3 dimensional input x=(x1,x2,x3) =(3, 2, 1) fully connected with weights (0.4, 0.2,0.4) to one neuron which is in the hidden layer. Calculate the output of the hidden layer neuron if the activation function is : (6)

(i) Sigmoid function

(ii) ReLU function.

14 a) Explain the Perceptron Algorithm. Implement AND function using perceptron networks for Bipolar inputs and Bipolar targets. (7)

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b) Why does a single perceptron cannot simulate a simple XOR function? Explain how to overcome this limitation. (7)

Module -3

15 a) Explain Deep Forward neural Network concept. (6)

b) Explain about any four regularization techniques in deep learning. (8)

16 a) Differentiate gradient descent with and without momentum. Give equations for weight updation in GD with and without momentum. Describe any 2 advantages of using Adam optimizer instead of basic gradient descent. (10)

b) State how to apply early stopping in the context of learning using Gradient Descent. Write the importance of validation set in early stopping. (4)

Module -4

17 a) Explain the different types of layers present in CNN with the purpose of each layer. (10)

b) Perform convolution to get a feature map of size 3*3. (4)

Input

3	3	2	1	0
0	0	1	3	1
3	1	2	2	3
2	0	0	2	2
2	1	0	3	1

Filter

0	1	2
2	2	0
0	1	2

18 Illustrate in detail the working of AlexNet architecture. (14)

Module -5

19 a) Draw and explain the architecture of GRU (10)

b) Why is the vanishing gradient problem is more pronounced in RNN than in traditional neural networks. Discuss a solution for the problem. (4)

20 a) Draw and explain the architecture of LSTM (10)

b) Explain how back propagation in RNN is different from that of normal feed forward NN. (4)

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